

Domain Adapted End to End Knob Tuner for Relational Databases

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Abstract

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Project Objectives

- Automate the manual trial-and-error suboptimal knob tuning process in relational databases.
- Engineer and develop a fully local tuning engine to optimize databases.

Related work

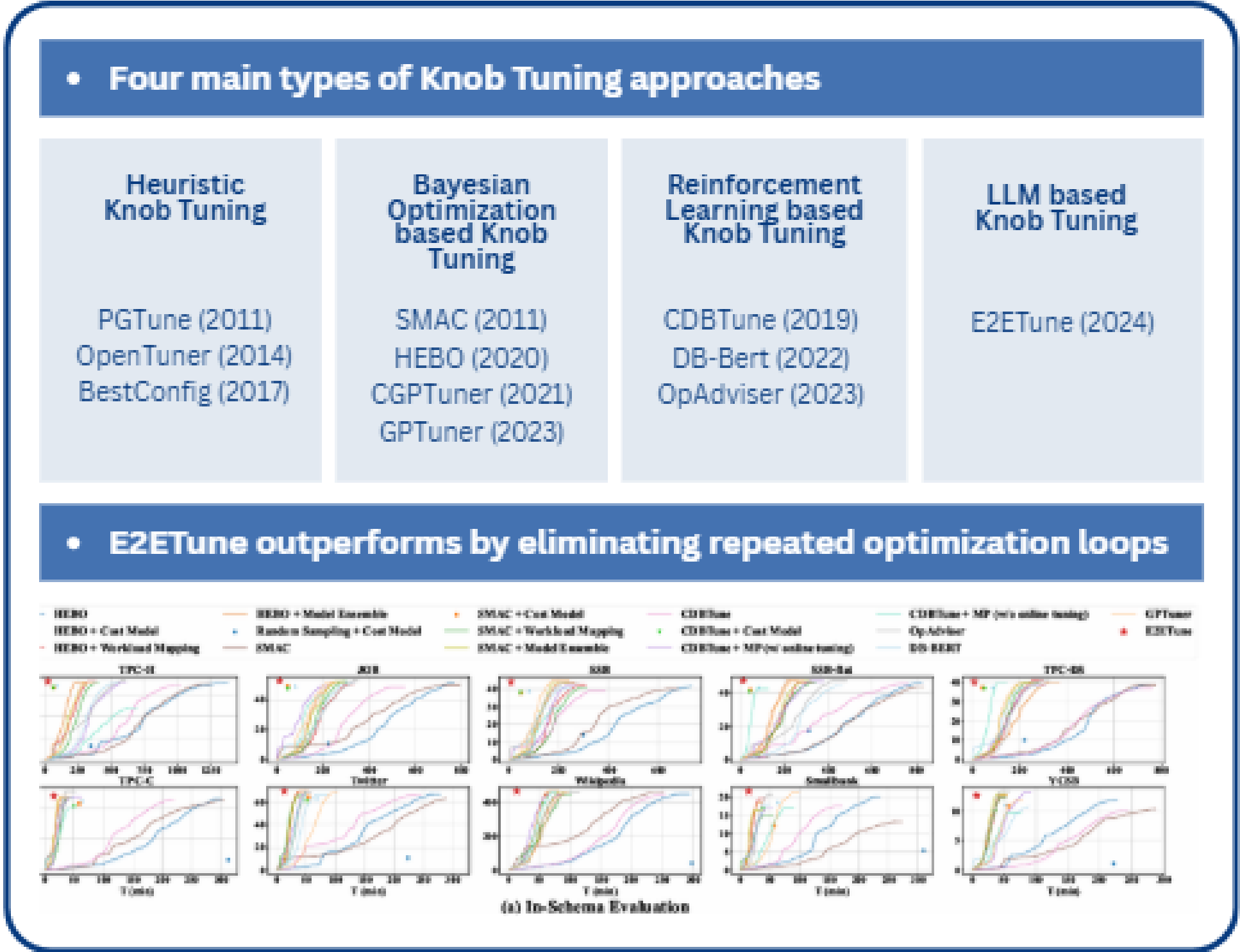
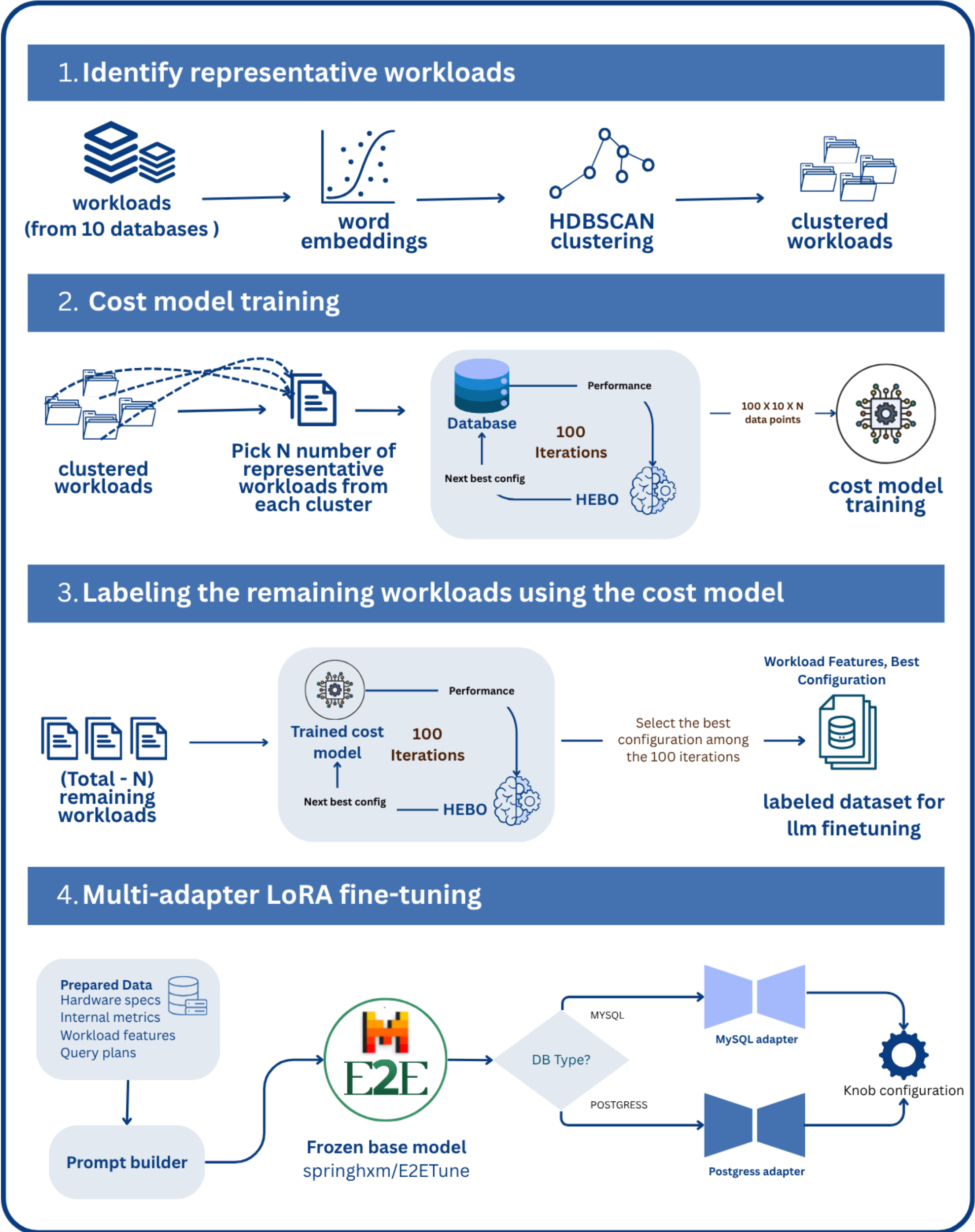


Figure 1. Related work overview and position of E2ETune.

Our work builds on E2ETune by adapting it to different database engines via domain adaptation.

Methodology

The requirement is that given a set of SQL queries, we want to find the best database configuration that optimizes the performance. We first collect data for 12 workloads, where each workload is a set of SQL queries, each from 10 different databases. This process is then repeated for PostgreSQL and MySQL on 2 different servers. This data is used train 2 cost models, one for each database which can be used to predict the cost of a workload regardless of server specs. Using this cost model we gather data points for more workloads. This way we have a large dataset of workloads, their features and the best database configuration for the workload.



Experiments

Cost Model

This model predicts workload latency and is used to accelerate the data-labeling process.

Approaches

- **MySQL-only Regressor:** Trained exclusively on MySQL samples with no source-domain transfer.
- **Residual Adapter:** Learns a residual correction over the frozen PostgreSQL base model predictions.
- **Joint Weighted Training:** Retrained jointly on PostgreSQL and MySQL data with domain-aware weighting.

5-Fold Cross-Validation Results

Model	RMSE	MAPE (%)	Spearman
MySQL-only Regressor	3.3853 ± 0.6655	12.26 ± 5.85	0.4693 ± 0.0437
Residual Adapter	3.3402 ± 0.6729	5.93 ± 0.73	0.5681 ± 0.0431
Joint Weighted Training	3.1431 ±0.5958	4.81 ± 0.23	0.6118 ± 0.0359

Table 1. Cost Model Evaluation Metrics

LLM Fine Tuning

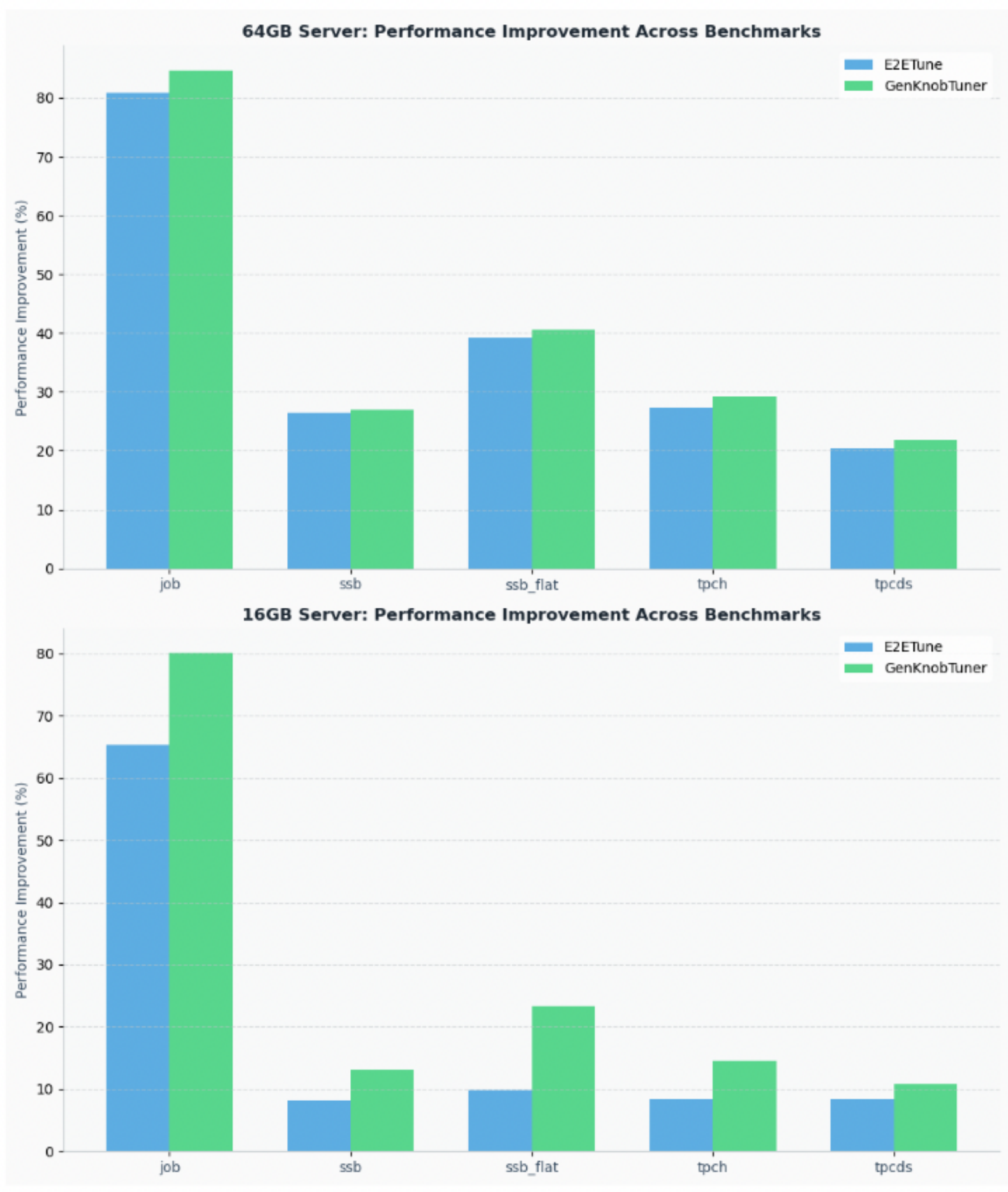
$$\frac{OptimizedTPS - DefaultTPS}{DefaultTPS} \times 100$$

Benchmark	SFT		Multi-adapter LoRA	
	MySQL	Postgres	MySQL	Postgres
job	n/a	84.66%	39.35%	n/a
ssb	n/a	27.02%	69.56%	n/a
ssb_flat	n/a	40.56%	40.32%	n/a
tpch	n/a	29.22%	47.46%	n/a
tpcds	n/a	21.84%	27.15%	n/a

Table 2. Performance improvement of average query latency over default configuration across benchmark queries on a 64GB, 4 core Intel Xeon E3-1275V6, with 512GB SSD.

Results

E2ETune vs GenKnobTuner Performance comparison on Postgres across different benchmarks



Performance improvement of GenKnobTuner on MySQL across different benchmarks

Benchmark	16GB	64GB
job	39.70%	39.35%
ssb	57.44%	69.56%
ssb_flat	24.03%	40.32%
tpch	44.78%	47.46%
tpcds	23.51%	27.15%

Conclusions

- Invenimus feles esse optima animalia domestica.
- Paene totum diem hoc exemplar faciendo impendi.
- Non satis pro hoc stipendium accipio.

References

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